

Needle Insertion Robot User Manual

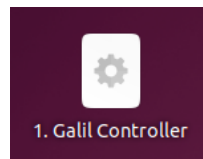
Version 1

Controller Setup

1. Login to the computer labeled “Robot”:
 - a. User: Yujie
 - b. Password: yujie312
2. Turn on the Galil Controller by pressing the black switch on the exterior of the controller box.
3. Turn off the Wifi (if it is on).
4. In the top right corner of the screen, an icon for “Wired Connecting” will appear once the Galil Controller is available for connection. Select “Internal Galil”.

Launching the GUI

1. Click on the Desktop icon labeled “1. Galil Controller”:



2. Wait for the controller GUI to appear before you move on to step 3. It is normal for a warning to appear regarding Wayland. However, make sure no other errors have been thrown. If there are any errors, make sure that Controller Setup steps 2-4 were all completed.
3. Click on the Desktop icon labeled “2. Robot GUI”:



Performing a Robotic Insertion

The main tab of the GUI contains 3 main buttons to perform an insertion experiment:

Insertion Experiment
Manual Control
Manual Homing

Insertion Experiment Instructions:

1. Select your needle length
2. Select the y-axis index for insertion (Select '1' to leave in home position)
3. Click 'Move to start' to bring robot to start position for experiment
4. Select the desired insertion increment
5. Click 'Insert Needle*' to insert the needle the amount specified in step 4
6. Repeat steps 4 and 5 until your needle insertion is complete
7. Click 'Remove needle**' to return to the start position specified in steps 1-2

*This will enable the Y/Z axis lock and prevent any movement in the Y/Z axes
**This will disable the Y/Z axis lock

Set Up Insertion

Select needle length

Select index

Move to start position

Perform Insertion

Select insertion type

Insert Needle

Finish Insertion

Remove Needle

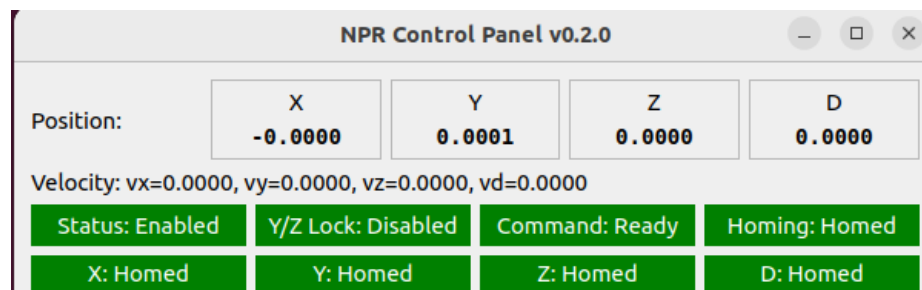
Detailed experiment instructions:

1. Choose the needle length (200 and 300mm needles are supported).
2. Choose the index: this is the position along the y-axis. This can be used to avoid inserting into the same spot in the phantom multiple times.
3. Click “Move to start position”. This will automatically move the robot the chosen distance away from the gel phantom so that the needle can be placed.
4. Attach the needle to the needle holder on the robot.
5. Ensure the tip of the needle is in the needle guide and that the z-axis is at the correct height for the needle guide. If it is not, please read the “Other GUI Features: Manual Control” section of this guide and then use the z-axis jog button align the needle to the guide.
6. Select the insertion type:
 - a. Full insertion: Will insert the needle to the maximum depth of the robot without stopping.
 - b. Incremental (5, 10, 20, or 50mm): Will insert the needle only by the amount specified and then stop.

- Click “Insert Needle”. This will move the robot to the end position, which is calculated based on the insertion type selected.
- Repeat steps 6-7 until insertion is complete. The “insertion type” can be changed during insertion (ex: begin with 5mm, then switch to 20mm, then fully insert).
- Once you would like to remove the needle, click on the “Remove Needle” button to return to the start position (determined by the values entered in steps 1 and 2, so be sure not to change them!).

Other GUI Features

Key Attributes



At the top of the control panel, the values of the following attributes will be displayed:

- Position: The current position of each axis with respect to the home position (in meters). The X, Y, Z, and D axes are labeled on the robot below:

TODO: ADD LABELED ROBOT PICTURE

- Velocity: The current velocity of each axis (in meters per second).
- Status: The status of the robot (will either be enabled or disabled). Must be enabled for the robot to move.
- Y/Z Lock: A blocker for the Y and Z axes. When locked, neither axis can be moved. This is automatically enabled when an insertion begins (“Insert Needle” is clicked) and automatically disabled when the insertion ends (“Remove Needle” is clicked).
- Command: Whether or not the robot is busy executing a command (either “Ready” or “Busy”).
- Homing: “Homed” if all 4 axes are at their correct home positions.
 - X, Y, Z, and D home status is shown beneath, to identify which axes specifically are homed.

Console Output

```
Console Output:
move_to_setpoint called
Moving to x=-0.0000, y=0.0034, z=0.0000, d=0.0000
move_to_setpoint called
Moving to x=-0.0000, y=0.0035, z=0.0000, d=0.0000
Insertion lock enabled: y and z axis movement is not permitted.
Insertion lock enabled: y and z axis movement is not permitted.
```

The output from commands will be shown in the console output section, including errors such as the robot not being enabled, or the robot being locked.

Manual Control

Insertion ExperimentManual ControlManual Homing

Jog Controls

↑

↑

↑

↑

X

Y

Z

D

↓

↓

↓

↓

Auto-Home (X,Y,Z)

Enabled

Unlocked

Set Position

X: X Setpoint [-0.05, 0.05]

Y: Y Setpoint [-0.05, 0.05]

Z: Z Setpoint [-0.05, 0.05]

D: D Setpoint [-0.07, 0.065]

Set Position

TODO: Add this section

Manual Homing

TODO: Add this section

Version History

Version	Date	Changes
1	August 16, 2025	Completed sections for controller setup, launching the GUI, and performing a robotic insertion.